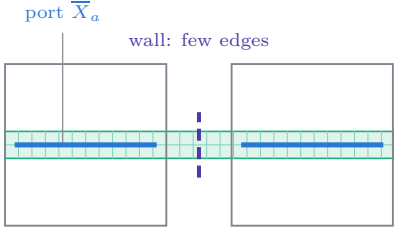
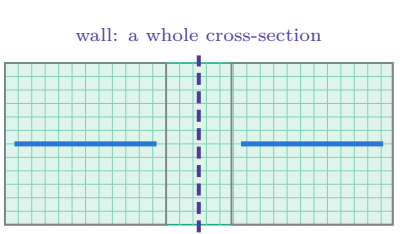
(a) Thin interface: a short wall



$X(\text{wall}) \simeq \overline{X}_a$ after the measurement,

so $\widetilde{d}_X \leq$ (edges in the wall)

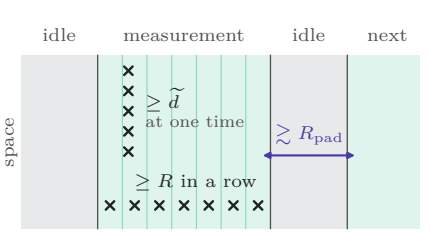
(b) Spanning grid: large separating cuts



$\Theta(N^2)$ edges, the scale of d;

m_0 ancillas per edge give $\widetilde{d}_X \geq d_X$

(c) Time: phenomenological faults



fault distance $\gtrsim \min\{\widetilde{d}, R, R_{\text{pad}}/\tau\}$